

ASHWIN SATISH

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EDUCATION

Northeastern University - Khoury College of Computer Science

Sept 2024 - Dec 2026 | Boston, MA

Master of Science, Data Science

GPA: 3.83/4.0

Machine Learning, NLP, MLOps, Algorithms, Data Mining, Database, Capstone

Christ University

June 2020 - May 2023 | Bangalore, India

Bachelor of Science, Computer Science

GPA: 3.53/4.0

EXPERIENCE

Gyori Lab for Computational Biomedicine | Machine Learning Developer Intern

Jan 2026 - Present | Boston, MA

- Engineered [TrialSynth](#)'s Trial Result LLM extraction pipeline in **Python** by parsing **35M** PubMed records, identifying **278K+ publications** linked to registered trials, transforming **13K+ papers** into structured data for clinical-trial knowledge graphs.
- Designed & built [CoGEx](#)'s Trial Result data model, transforming trial datasets to a **Neo4j graph**, generating **115K+ nodes** & **116K+** relationships with **REST APIs** & [Flask search interface](#) enabling queryable clinical trials within the knowledge graph.
- Architected a custom grounding pipeline for Trial Result data by integrating the Ontology of Adverse Events (OAE), improving adverse events grounding coverage from **75.1% to 85.4%**.
- Benchmarked LLMs on **AWS Bedrock** for clinical-data extraction across cost, recall, and token efficiency, cutting per-paper inference cost **~65%** while matching **90% recall** against the baseline model.

Northeastern University | Machine Learning Researcher

Jan 2025 - Dec 2025 | Boston, MA

- Boosted true positives from **2% to 77%** (maintaining zero false positives) for Lyme cell detection via 7-model CNN ensemble.
- Built a **Python** image preprocessing pipeline for **30,000+** MATLAB cell images, saving the biology team weeks of manual work.
- Improved accuracy by **13%** via augmentation & L2/dropout regularization, achieving production-ready performance.
- Mentored & conducted office hours as a Teaching Assistant for **300+ students** in linear algebra, statistics & **NumPy** demos.

CarerAI | Data Science Intern

Jul 2023 - May 2024 | Gurgaon, India

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through KNN analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's score by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

SKILLS

- Languages:** Python, SQL, R, C, Java, HTML, CSS, TypeScript, JavaScript
- Libraries & Frameworks:** Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, OpenCV, Pandas, NumPy, PySpark
- Cloud & Development:** AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j, Bedrock, Lambda, Kubeflow
- LLM & GenAI Tooling:** Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Ollama, PydanticAI, MLFlow

PROJECTS

AI-Powered Pokémon Battle Strategy Engine | *Ollama, ChromaDB, Sentence-Transformers, Docker, Streamlit* | [Github](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating Llama 3.2 with deterministic backend for strategy analysis with a recommendation system for optimal team synergies by ingesting **1M+** aggregated usage **stats** via Python **ETL** pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, indexing **40,000+** strategy docs to ground LLM inference.

PantryPilot: Pantry & Recipe Assistant (2nd, MLOps Expo: Google) | *LoRA, PostgreSQL, Airflow, Docker, GCP, Ollama* | [Github](#)

- Contributed to the fine tuning, evaluation & automated deployment (CI/CD) on **GCP** of Llama 3.2 3B for personalized inventory recipes, resulting in **50% reduction** in dietary violations via LoRA adaptation.
- Collaborated on **DPO feedback** with Teacher model (Llama 3 70B) for preference labelling, VLM for OCR & Veo 3.1 for video.

Airlytics for Fitbit: Analysis, Forecast & Health Coach | *FastAPI, SQLAlchemy, SQLite, Groq, Scikit-learn, Pandas* | [Github](#)

- Built personalized readiness-forecasting models & dashboards for Fitbit Air's data, **beating a naive baseline** by **16%**.
- Built LLM health coach that provides health insights on retrieved data, validated by scoring tool-call accuracy & grounding rate.

arXiv Research Assistant: Agentic RAG for Q&A | *OpenSearch, LangChain, LiteLLM, NetworkX, Weights & Biases* | [Github](#)

- Built an agentic RAG assistant answering questions over arXiv papers via hybrid BM25/kNN retrieval and a GraphRAG graph.
- Engineered a semantic chunker & ingestion pipeline, indexing 1785 chunks from 10 arXiv papers in under 30s (79/s).